Project 3 Harmonic Analysis

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1 Introduction

Definition 1 (Weights)**.** *A* weight *is a function $w \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$ with $w \geq 0$. For a measurable set $E \subseteq \mathbb{R}^d$ we write $w(E) := \int_E w(x)\,dx$.*

Definition 2 (Weighted L^p spaces)**.** *Let w be a weight and let $1 < p < \infty$. We define*

$$L^p_w(\mathbb{R}^d) := \left\{ f : \mathbb{R}^d \to \mathbb{R} \text{ measurable} : \int_{\mathbb{R}^d} |f(x)|^p\, w(x)\,dx < \infty \right\},$$

with norm $\|f\|_{L^p_w} := \left(\int_{\mathbb{R}^d} |f(x)|^p\, w(x)\,dx\right)^{1/p}$.

Definition 3 (Hardy–Littlewood maximal operator)**.** *For $f \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$ and $x \in \mathbb{R}^d$,*

$$Mf(x) := \sup_{Q \ni x} \frac{1}{|Q|} \int_Q |f(y)|\,dy, \tag{1}$$

where the supremum is taken over all cubes $Q \subset \mathbb{R}^d$ with sides parallel to the axes containing x. If the supremum is taken only over cubes Q centred at x, we write M^c.

It is well known that M is of weak type $(1,1)$ and bounded on $L^p(\mathbb{R}^d)$ for $1 < p \leq \infty$. A natural question is therefore, for fixed $1 < p < \infty$, for which weights w is the operator M bounded on $L^p_w(\mathbb{R}^d)$? In other words, for which w is there a constant C such that for all $f \in L^p_w(\mathbb{R}^d)$

$$\|Mf\|_{L^p_w} = \left(\int_{\mathbb{R}^d} (Mf)^p\, w\,dx\right)^{1/p} \leq C \left(\int_{\mathbb{R}^d} |f|^p\, w\,dx\right)^{1/p} = C\,\|f\|_{L^p_w}. \tag{2}$$

Remark 1 (The zero weight)**.** *If $w = 0$ almost everywhere, then $\|f\|_{L^p_w} = 0$ for every f, so $L^p_w(\mathbb{R}^d)$ is trivial and M is trivially bounded.*

Remark 2 (The endpoint $p = \infty$)**.** *For $p = \infty$, with the essential supremum taken with respect to $w(x)\,dx$, the answer is trivial: M is bounded if and only if $w > 0$ almost everywhere. If $w > 0$ a.e., then $w(x)\,dx$ and dx have the same null sets, so $\|Mf\|_{L^\infty} \leq \|f\|_{L^\infty}$ applies. If $w = 0$ on a set Z with $|Z| > 0$, then $f = \chi_Z$ has weighted supremum 0, while $Mf > 0$ everywhere. Hence we restrict to $1 < p < \infty$.*

Lemma 1. *Let $1 < p < \infty$ and let w be a weight that is not almost everywhere zero. If M is bounded on $L^p_w(\mathbb{R}^d)$, then $w > 0$ almost everywhere.*

Proof. Suppose, for contradiction, that $w = 0$ on a measurable set E with $|E| > 0$. Since $|E| = \lim_{R\to\infty} |E \cap [-R, R]^d|$, there is a bounded measurable set $E' \subseteq E$ with $|E'| > 0$. Let $f := \chi_{E'}$. Then $f \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$, and since $w = 0$ on E',

$$\|f\|^p_{L^p_w} = \int_{E'} w(x)\,dx = 0.$$

On the other hand, let $x \in \mathbb{R}^d$. Since E' is bounded, there is a cube Q containing both x and E', and then

$$Mf(x) \geq \frac{1}{|Q|}\int_Q \chi_{E'}(y)\,dy = \frac{|E'|}{|Q|} > 0.$$

Hence $Mf > 0$ everywhere. Since w is not almost everywhere zero, the set $\{w > 0\}$ has positive measure, and $(Mf)^p w > 0$ on it. Therefore $\|Mf\|_{L^p_w} > 0 = C\,\|f\|_{L^p_w}$, which contradicts (2). □

Definition 4 (Muckenhoupt A_p weights)**.** *Let $1 < p < \infty$ and let w be a weight. The A_p* characteristic *of w is*

$$[w]_{A_p} := \sup_Q \left(\frac{1}{|Q|}\int_Q w(x)\,dx\right)\left(\frac{1}{|Q|}\int_Q w(x)^{-\frac{1}{p-1}}\,dx\right)^{p-1}, \tag{3}$$

where the supremum is taken over all cubes $Q \subset \mathbb{R}^d$. We say that $w \in A_p$ if $[w]_{A_p} < \infty$.

References